

SPIP Year 7 Science Pre-test Answer Key

Subject: Science Level: SPIP Year 7 Total: 50 marks Generated from system answer keys: 2026-06-24

Questions 1-4

1. spip-y7s-q1 3 marks

Question 1: Some materials are electrical conductors and others are electrical insulators. Complete the table about these materials.

Answer: copper: conductor; graphite: conductor; plastic, rubber, wood: insulator

copper (0.6 mark): accepted: conductor

graphite (0.6 mark): accepted: conductor

plastic (0.6 mark): accepted: insulator

rubber (0.6 mark): accepted: insulator

wood (0.6 mark): accepted: insulator

Threshold scoring: 5 correct = 3 mark(s); 3 correct = 2 mark(s); 1 correct = 1 mark(s)

2. spip-y7s-q2 3 marks

Question 2: Label the organs on the diagram of a human body.

Answer: brain; heart; lung; stomach; kidney; small intestine

brain (0.5 mark): accepted: brain

heart (0.5 mark): accepted: heart

kidney (0.5 mark): accepted: kidney, kidneys

lungs (0.5 mark): accepted: lung, lungs

stomach (0.5 mark): accepted: stomach

intestines (0.5 mark): accepted: intestine, intestines, small intestine

Threshold scoring: 6 correct = 3 mark(s); 4 correct = 2 mark(s); 2 correct = 1 mark(s)

3. spip-y7s-q3 1 mark

Question 3: Mike is exploring electrical circuits. The lamps are very dim. What can he do to make the lamps brighter?

Answer: add another cell

answer (1 mark): accepted: add another cell

4. spip-y7s-q4a 1 mark

Question 4(a): Blessy keeps the temperature of the water for each solid the same. Explain why.

Answer: So it is a fair test because temperature affects how much solid dissolves.

Status: reviewRecommended

answer (1 mark): keywords: fair + test OR temperature + affect + dissolv OR same + compare OR hotter + dissolv OR colder + dissolv OR different + amount; teacher review recommended

5. spip-y7s-q4b 1 mark

Question 4(b): Blessy thinks it is a good idea to repeat her investigation. Explain why.

Answer: Repeating makes the results more reliable and helps check for mistakes.

Status: reviewRecommended

answer (1 mark): keywords: reliable OR accurate OR check + mistake OR average; teacher review recommended

6. spip-y7s-q4c 3 marks

Question 4(c): Blessy has started to draw a bar chart of the results. Complete the bar chart. Include the scale on the y-axis, label on the y-axis, and the other three bars and their labels.

Answer: scale 10, 20, 30; fertiliser 30 g; salt 8 g; baking powder 5 g

Status: reviewRecommended

scale_top (0.25 mark): accepted: 30, 30 g

scale_middle (0.25 mark): accepted: 20, 20 g

scale_bottom (0.25 mark): accepted: 10, 10 g

solid_4 (0.25 mark): accepted: fertiliser, fertilizer

fertiliser (0.5 mark): accepted: 30, 30 g

solid_5 (0.25 mark): accepted: salt

salt (0.5 mark): accepted: 8, 8 g

solid_6 (0.25 mark): accepted: baking powder

baking_powder (0.5 mark): accepted: 5, 5 g

7. spip-y7s-q4d 1 mark

Question 4(d): Which solid is the most soluble in water?

Answer: fertiliser

answer (1 mark): accepted: fertiliser, fertilizer

Questions 5-8

8. spip-y7s-q5a 2 marks

Question 5(a): Use the information to draw a food chain. Draw arrows between the boxes to show the direction of energy flow.

Answer: leaf -> caterpillar -> bird -> snake -> owl

pos1 (0.4 mark): accepted: leaf

pos2 (0.4 mark): accepted: caterpillar

pos3 (0.4 mark): accepted: bird

pos4 (0.4 mark): accepted: snake

pos5 (0.4 mark): accepted: owl

9. spip-y7s-q5b 1 mark

Question 5(b): Which living thing is the producer in this food chain?

Answer: leaf

answer (1 mark): accepted: leaf, plant, the leaf, tree

10. spip-y7s-q6a 1 mark

Question 6(a): Yuri wants to separate a mixture of salt, sand and water. What dissolves in stage B?

Answer: salt

answer (1 mark): accepted: salt

11. spip-y7s-q6b 1 mark

Question 6(b): What is substance X?

Answer: salt

answer (1 mark): accepted: salt, wet salt

12. spip-y7s-q7a 3 marks

Question 7(a): Complete Lily's sentences. Choose from: centimetres, kilograms, newtons, seconds.

Answer: Mass: kilograms; Weight: newtons; Force: newtons

mass (1 mark): accepted: kilograms, kg

weight (1 mark): accepted: newtons, newton

force (1 mark): accepted: newtons, newton

13. spip-y7s-q7b 1 mark

Question 7(b): Choose the arrow that shows the direction of the force of gravity on Lily.

Answer: down

answer (1 mark): accepted: down

14. spip-y7s-q8a 1 mark

Question 8(a): Only one solid can be separated from water by filtration. Which one?

Answer: C

answer (1 mark): accepted: C

15. spip-y7s-q8b 1 mark

Question 8(b): There is a reversible change when solid A is added to water. Describe how you could reverse this change.

Answer: Heat it, evaporate the water, or leave it in the Sun to get the solid back.

Status: reviewRecommended

answer (1 mark): keywords: evaporat OR heat OR sun OR water + solid OR crystal; teacher review recommended

16. spip-y7s-q8c 2 marks

Question 8(c): Two of the solids have an irreversible change when added to water. Write the letter of one of these solids. Explain how you can tell from the results.

Answer: D because it fizzes/forms gas, or E because the temperature changes/gets colder.

Status: reviewRecommended

letter (1 mark): accepted: D, E

explanation (1 mark): keywords: fizz OR gas OR bubble OR colder OR temperature + change; teacher review recommended

Questions 9-12

17. spip-y7s-q9a 2 marks

Question 9(a): Mike must use the heart machine to stay alive. Explain what the heart machine does.

Answer: It does the work of the heart: it pumps blood around the body, supplying oxygen and food/nutrients to organs.

Status: reviewRecommended

answer (1 mark): keywords: does + heart OR place + heart OR instead + heart OR takes + heart OR heart + normally; teacher review recommended

answer (1 mark): keywords: pump + blood OR circulat + blood; teacher review recommended

answer (1 mark): keywords: oxygen; teacher review recommended

answer (1 mark): keywords: food OR nutrient; teacher review recommended

answer (1 mark): keywords: organ OR around + body OR body; teacher review recommended

18. spip-y7s-q9b 1 mark

Question 9(b): Mike takes the extra heart machine with him when he goes outside. Explain why Mike needs an extra heart machine.

Answer: In case the first heart machine or its batteries stop working or break.

Status: reviewRecommended

answer (1 mark): keywords: break OR stop + work OR fail OR backup OR spare OR battery; teacher review recommended

19. spip-y7s-q10 1 mark

Question 10: Draw a line from the statement to the correct explanation.

Answer: A rollercoaster is able to climb up the hill because its movement gives it the energy to get to the top of the hill.

climb (1 mark): accepted: movement energy

20. spip-y7s-q11a 1 mark

Question 11(a): Glass, plastic and metal can be recycled. Write down the name of another material that can be recycled.

Answer: paper, card/cardboard, cloth, books, magazines, batteries, or ink cartridges

answer (1 mark): accepted: paper, cardboard, card, cloth, book, books, magazine, magazines, clothes, battery, batteries, ink cartridge, ink cartridges

21. spip-y7s-q11b 2 marks

Question 11(b): Complete the sentences about why the diaper cannot be recycled and how to reduce waste in the environment.

Answer: It is dirty, toxic, or contains microbes; use a washable/reusable diaper, or compost a biodegradable one.

Status: reviewRecommended

cannot_recycle (1 mark): keywords: dirty OR waste OR soiled OR toxic OR microbe OR mixed + material OR cannot + recycle OR used; teacher review recommended

reduce_waste (1 mark): keywords: reuse OR wash OR cloth OR reusable OR compost OR biodegrad; teacher review recommended

22. spip-y7s-q12a 2 marks

Question 12(a): Complete the sentences about sugar water. Choose from: insoluble, soluble, solution, sugar, water.

Answer: solvent: water; solute: sugar; sugar is soluble

solvent (0.67 mark): accepted: water

solute (0.67 mark): accepted: sugar, soluble

soluble (0.66 mark): accepted: soluble

23. spip-y7s-q12b 1 mark

Question 12(b): When sugar dissolves in water, is the sugar still in the water?

Answer: yes

answer (1 mark): accepted: yes

Questions 13-16

24. spip-y7s-q13a 2 marks

Question 13(a): Pierre is testing which materials are electrical conductors. Put each instruction letter in the correct order.

Answer: D, A, C, B

pos1 (0.5 mark): accepted: D

pos2 (0.5 mark): accepted: A

pos3 (0.5 mark): accepted: C

pos4 (0.5 mark): accepted: B

25. spip-y7s-q13b 1 mark

Question 13(b): Pierre thinks one of his results is incorrect. He wants to test this material again. Which material does he test again?

Answer: steel

answer (1 mark): accepted: steel

26. spip-y7s-q13c 1 mark

Question 13(c): Pierre makes a conclusion from his results. What conclusion does Pierre make?

Answer: Metals conduct electricity and non-metals do not.

Status: reviewRecommended

answer (1 mark): keywords: metal + conduct OR non + metal + not OR plastic + stone + not OR iron + lead + copper; teacher review recommended

27. spip-y7s-q14a 1 mark

Question 14(a): Aiko measures the volume of water collected from each tap. Write down the name of the apparatus she uses.

Answer: measuring cylinder

answer (1 mark): accepted: measuring cylinder, measuring cylinders, graduated cylinder

28. spip-y7s-q14b 2 marks

Question 14(b): Aiko writes down the results. Complete her table of results.

Answer: tap 1: 0.0 cm³; tap 2: 1.8 cm³; tap 3: 2.9 cm³; tap 4: 3.8 cm³; tap 5: 3.3 cm³

tap1_volume (0.33 mark): accepted: 0.0, 0, 0.0 cm³, 0 cm³

tap2_number (0.33 mark): accepted: 2, tap 2

tap3_volume (0.33 mark): accepted: 2.9, 2.9 cm³

tap4_number (0.33 mark): accepted: 4, tap 4

tap4_volume (0.34 mark): accepted: 3.8, 3.8 cm³

tap5_volume (0.34 mark): accepted: 3.3, 3.3 cm³

Threshold scoring: 6 correct = 2 mark(s); 4 correct = 1 mark(s)

29. spip-y7s-q14c 1 mark

Question 14(c): There are drips from all the taps. One of the results is wrong. Circle the result that is wrong. Explain your answer.

Answer: tap 1, because all taps drip but tap 1 has 0.0 cm³/no volume of water collected.

Status: reviewRecommended

choice (0.5 mark): accepted: tap 1

explanation (0.5 mark): keywords: all + tap + drip OR 0 OR no + water OR no + volume OR nothing + collected OR other + volume; teacher review recommended

30. spip-y7s-q15a 1 mark

Question 15(a): Complete the sentence: When the toy bounces up, the upward force is _____ than the downward force.

Answer: greater

answer (1 mark): accepted: greater, larger, more, bigger, stronger

31. spip-y7s-q15b 1 mark

Question 15(b): What does Jamila do to make the toy bounce faster?

Answer: push on the spring more often

answer (1 mark): accepted: push on the spring more often

32. spip-y7s-q16a 1 mark

Question 16(a): Chen cannot use a sieve to separate the mixture of sand and copper sulfate. Explain why.

Answer: The particles are too small or similar size, so both solids would pass through the sieve.

Status: reviewRecommended

answer (1 mark): keywords: particle + small OR same + size OR both + pass OR powder; teacher review recommended

33. spip-y7s-q16b 1 mark

Question 16(b): What substance does the residue contain?

Answer: sand

answer (1 mark): accepted: sand

34. spip-y7s-q16c 1 mark

Question 16(c): What is the name of the filtrate?

Answer: copper sulfate solution

answer (1 mark): accepted: copper sulfate solution, copper sulphate solution, copper sulfate, copper sulphate, blue solution

35. spip-y7s-q16d 1 mark

Question 16(d): What colour is the filtrate?

Answer: blue

answer (1 mark): accepted: blue
